Joshua Si

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EDUCATION

McMaster University Hamilton, Ontario
Bachelor of Engineering – Computer Engineering *August 2025 – June 2029*

- Received $3000.00 Faculty of Engineering Award of Excellence Entrance Scholarship
- Relevant Coursework: Digital Logic Design (Quartus, FPGA), Circuit Analysis
- Built and tested circuits with an **Analog Discovery 3**, verifying nodal and mesh analysis against oscilloscope and voltmeter readings

EXPERIENCE

Backend/Full-Stack Developer Jul 2026 – Present
McMaster Engineering Society *Hamilton, Ontario · Remote*

- Resolved bugs and shipped features across **4 web platforms** used by **thousands** of McMaster Engineering students, built with **Next.js**, **TypeScript**, and **MongoDB**
- Debugged, tested, and maintained the **1,174-file codebase** behind a **university-wide website**, keeping it reliable, fast, and current
- Collaborated with a cross-functional team in **GitHub** and **Figma** through issue tracking, code reviews, testing, and deployment

Sponsorship Coordinator Jul 2026 – Present
Google Developer Groups McMaster University *Hamilton, Ontario · Hybrid*

- Secured **$1100+** in funding and in-kind assets for a hackathon by negotiating with companies including **Bell**, **Rogers**, and **Cloudinary**
- Automated sponsor outreach for a **30-person team** with an agentic AI workflow, reaching **100+ sponsors** and saving **10+ hours**

Project Administrator September 2025 – April 2026
McMaster University *Hamilton, Ontario*

- Developed and integrated a **458-line Python** control program with a 3D-printed robotic end effector, improving grip effectiveness by **40%**
- Gathered client requirements and translated them into a technical design, coordinating a **5-person team** on a **12-week timeline**

Private Tutor April 2024 – August 2024
Self-Employed *Toronto, Ontario*

- Improved academic performance by **10 to 20%** across **10+ students** in weekly **2-hour** sessions in high school Calculus, Physics, and Java

PROJECTS

ArtMentor AI | JavaScript, React, Gemini API, Express February 2026

- Collaborated on an **AI-powered** application that analyzes user-submitted artwork and returns visual and textual feedback for **100+ users**
- Integrated native APIs, responsive UI patterns, and full-stack JavaScript across **3+ device form factors**, reducing page load time by **35%**
- Fulfilled **UN SDG 4** project requirements with a technology-driven solution focused on access to art education

MediExplain | Gemini API, JavaScript, Node.js, React March 2026

- Conceptualized an **AI-powered** medical literacy tool that bridges language and knowledge gaps in Canadian healthcare
- Led **Gemini API** integration for the tool, explaining medical terms across **5+ languages** in under **2 seconds**
- Built the front-end in **React** with **10+ interactive UI elements** and configured a **Node.js** backend with **300+ lines** of server-side logic

DianaWebsite | TypeScript, Next.js, Node.js, Express August 2026

- Built a **full-stack** application with a **REST API** of **10 routes**, **JWT** session authentication, and file uploads on one custom server
- Created drag-and-drop image uploads and in-place text editing so the owner can update content without touching code
- Secured **5 owner-only** routes behind **JWT** cookies and persisted content and image uploads to disk across restarts

Low Voltage System | EasyEDA Pro, PCB Layout, Schematic Capture, CAN Bus October 2026

- Designed and routed a **2-layer PCB** in **EasyEDA Pro** that distributes fused **12 V** power, **CAN**, and **Gigabit Ethernet** to the controllers and sensors of an autonomous vehicle
- Mapped the data flow of **7 devices**, including a **LiDAR** and a **PoE camera**, specifying each protocol (**CAN** at **500 kbps**, **TCP/UDP** over Ethernet, **PWM**) and destination
- Sized power traces to each branch fuse rating, from **32 mil at 2 A** to **110 mil at 5 A**, with ground pours and a clean design rule check
- Specified fuses and wire gauges for a **76 A** worst-case load, and costed a **14-item** bill of materials at **$4,211 CAD** on a **3-person team**

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, SQL, HTML, CSS
Software: React, Next.js, Node.js, Express, MongoDB, REST APIs, JWT Authentication, Git, GitHub, Linux, Gemini API
Hardware & Embedded: PCB Design, Schematic Capture, CAN Bus, Ethernet (TCP/UDP), PWM, Arduino, Raspberry Pi, FPGA (DE10-Lite)
Tools: EasyEDA Pro, Quartus, Analog Discovery 3, Autodesk Inventor, Figma, Cursor, Google Cloud Platform, Microsoft Excel, PowerPoint
Languages Spoken: English, Mandarin
Certificates & Awards: Generative AI: Working with Large Language Models; Agentic AI: Build Your First Agentic AI System; McMaster Thode Makerspace Certification; FIT Certificate for Software Design and Development; Ontario Scholar; Waterloo Math Distinction